

Global Facilities - Design & Construction Standard - Bulk Gas - Gas Plant (N₂, O₂, Ar) Design System

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1.0 Purpose

- 1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Gas Plant as part of new Greenfield FAB Construction projects
- 1.2 The standard is applicable for Gas Plant generating Nitrogen (N₂), Oxygen (O₂), and Argon (Ar) systems.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

- 2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.
- 2.2 Site(s) impacted
This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

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3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
ASU	Air Separation Unit
CDA	Clean Dry Air
CHW	Chilled Water
CO2	Carbon Dioxide
CQC	Continuous Quality Control
CW	Condenser Water
FMCS/SCADA	Facility Monitoring and Control System/ Supervisory Control and Data Acquisition
FM	Factory Mutual
ICA	Instrument Control Air
LAR	Liquid Argon
LIN	Liquid Nitrogen
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
LOX	Liquid Oxygen
MAC	Main Air Compressor
PLC	Programmable Logic Controller
PPB	Part per billion
PPM	Part per million
PPU	Pre-Purification Unit
TUM	Tool Utility Matrix
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

Roles	Responsibility
GFTT Lead (Discipline Engineer)	- Develop and maintain System Standard document - Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard - Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	- Review and provide technical feedback on System Standard - Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	- Review and provide technical feedback on System Standard - Incorporate System Standard into Project Engineering, Design, and Construction - Document project deviations, exceptions, and exclusions from System Standard - Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	Use System Standard as technical reference

5.0 Overview of Systems

5.1 Types of gas plant

On site gas generators are by far the most common type of Air Separation Plant. These plants can produce

- Oxygen only,
- Oxygen and Nitrogen,
- Oxygen, Nitrogen and Argon, or
- Nitrogen only

In all cases the gases that are not fully purified are used for refrigeration and vented to atmosphere. These units separate the air in liquid form but utilize the refrigeration from the liquids before the gases are vented.

Figure 1 shows the basic equipment and flow arrangement for a modern air separation plant. Air is compressed in the Main Air Compressor (MAC) to a nominal 100 psig or higher. It is then cooled to ambient temperature and passed through the Pre-Purification Unit (PPU). This consists of a pair of vessels containing a fixed bed of adsorbent, typically a combination of activated alumina and molecular sieve material.

As the air passes over the adsorbent, many of the trace contaminants are removed, especially water, carbon dioxide, and the heavy hydrocarbons. The purified air then enters the main heat exchanger, where it is cooled to near its liquefaction temperature before entering the distillation column. Oxygen is the highest boiling of the three main components, so it is taken from the bottom of the low-pressure column. Nitrogen is taken from the top of the low-pressure column. Argon splits between the oxygen and nitrogen and can be recovered as a pure product by adding a third distillation column. The product streams are warmed to ambient temperature against incoming air to recover the refrigeration. It is also possible to remove the products from the distillation system as liquid if sufficient refrigeration is provided.

Figure 1 also shows four possible locations of contaminant adsorbers which may or may not appear in a particular plant depending upon the design requirements.

With the distillation process, there is possibility of hydrocarbon build up in the O₂ which can cause an explosion. Some designs pull for O₂ distillation above the Crude O₂ liquid to avoid this issue. It is critical that the N₂ plant take this into account to mitigate the risk.

This standard will serve as a minimum requirement and guidelines. Specific details or requirement need to be addressed during detailed design for specific fab. Power consumption per Nm³ gas produced by the gas plant to be reviewed and benchmarked in-line with Micron's sustainability efforts in reducing energy consumption and carbon footprint.

Power consumption and metering need to be included and reporting to Micron on monthly, quarterly or yearly basis for performance monitoring and auditing.

Any other impact such as emissions, waste water, recycle effort to be included in the design and reported to Micron as well.

Preliminary design and guidelines shall be submitted to (Micron Risk Management Group) for review with Factory Mutual (FM). Recommendations and requirement by the team shall be evaluated and included during the design phase.

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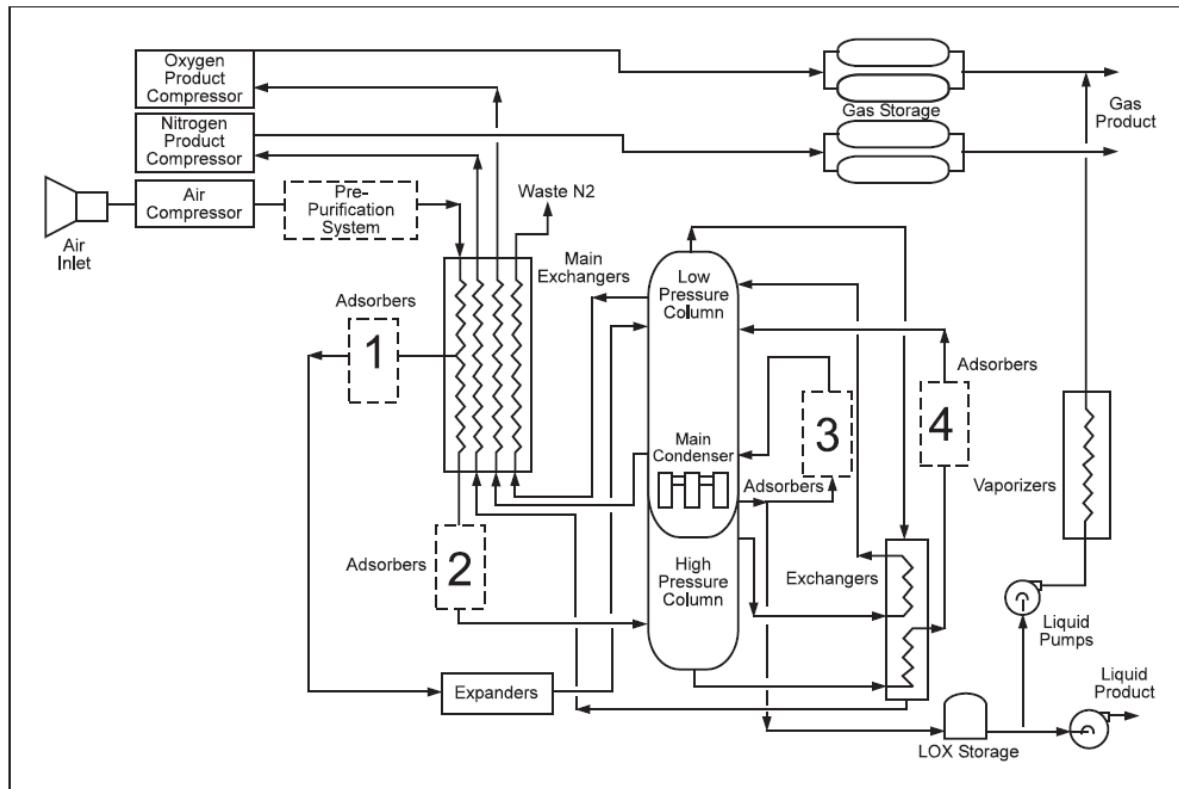


Fig. 1. Generic air separation plant flow diagram. Adsorbers labeled 1 through 4 may or may not appear in a particular plant depending on design requirements. (Source: CGA P-8)

5.1.1 All Liquid Product Plants

Generally, these plants may make liquid Oxygen (LOX), liquid Nitrogen (LIN) and liquid Argon (LAR). These liquid products are delivered into cryogenic tanks, where it is either warmed back to a gas before use or used as a liquid.

All Liquid Product Air Separation Plants may also have an additional piece of equipment called a Liquefier to provide the necessary refrigeration to liquefy all products. The Liquefier is generally the largest power user by far in an Air Separation Plant.

5.1.2 Combined Liquid and Gas Product Plants

Commonly called "piggyback plants," these Air Separation Plants produce mostly gaseous product for pipeline use, but also have Liquefiers to produce liquid products for storage in cryogenic tanks for use as back-up supply.

Gas suppliers shall provide following documentation to Micron for record keeping and future reference. Any changes to the document to be updated and maintained keeping Micron the updated document: Gas plant(s) description, system overview, capacity, and features

- i. Gas yard layout, plant footprint, and floor load calculation
- ii. Gas plant diagram blocks and P&ID
- iii. Gas plant preventive maintenance schedule
- iv. Business Contingency Plan and escalation mapping
- v. Part Number and detection limit of CQC analyzers

5.2 Gas specification and requirement

The performance requirements, as listed below, are defined at the boundary of the gas yard:

Gas Product from ASU	Delivery Pressure (at gas yard boundary)
Nitrogen	≥ 113 psig (7.8 barg)
Oxygen	≥ 113 psig (7.8 barg)
Argon	≥ 113 psig (7.8 barg)

As the pressure will be limited by MAC; and affecting power consumption, the requirement need to be reviewed in the design stage to have the most efficient configuration for manufacturing.

In determining the delivery pressure at the gas yard boundary, the gas plant supplier shall consider the expected pressure drop due to

downstream piping and equipment e.g. filter, purifier, distribution piping, etc. so that delivery pressure at the end of line valve is:

Gas Product from ASU	End of Line Pressure (Fab distribution)
Nitrogen	≥ 105 psig (7.2 barg)
General Oxygen	≥ 86 psig (5.9 barg)
General Argon	≥ 97 psig (6.7 barg)

The purity of the gaseous and/or liquid products produced by the air separation plant will be determined by the gas plant supplier with Micron's agreement. The purity of the bulk gases as delivered to the facility and fab must meet or exceed the following specifications. Refer to [Global Facilities – General – Design Parameters Best Known Methods \(BKM\)](#) for the latest updated required specifications.

Utility N2 purity may be produced directly from the air separation plant. However, PN2, PO2, and PAr purity specifications may be produced directly from the high purity air separation plant or require purifiers to meet Micron's product quality requirements. Evaluation need to be access based on gas supplier technology capability, risk and cost benefit to Micron in either options.

5.3 Main component description and requirement

5.3.1 Filter

- i. Upstream Pre-Filters are used before the main air compressor to remove dust and particulates. Most pre-filters are large fiberglass filters that do not present a fire hazard.
- ii. Post gas plant filter will be required for all bulk gases before supplying to production/fab. (covered in "Bulk Gas Delivery and Dispense Systems Design Guide")
- iii. Inlet air filters shall be sized to accommodate the maximum capacity of the total number of main air compressors that may be on-line at the same time at full demand/forecast. The design of the inlet air manifold should allow for installation of additional air filters in phases as additional MACs may be added to increase plant capacity to full buildout. Gas supplier shall select the particle filtration size suitable for operation with their main air compressors. Air filter shall be made of non-combustible material so as not to present a fire hazard.

- iv. The supplier shall also consider the following in selection of the inlet air filters:
 - Manufacturer's standard size offering the lowest capital cost.
 - Existing inlet air filters on site or nearest Micron fab to allow for sharing of spare filters.
 - Design must allow for regular replacement of air filters.
- v. Filters may be necessary to achieve the performance specifications as required by Micron. The total filter flow capacity shall be 25% larger than maximum capacity required from the ASU not including the spare "+1" filter. Filters must retain particles of 0.05 µm size at a minimum with no more than a 2 psi maximum pressure loss. The filter manifold shall be designed to allow installation of additional filters without interruption to the fab, and N+1 filters shall be maintained at all times if filters are added in phases.

5.3.2 Main Air Compressor

- i. The MAC compresses atmospheric air, delivers it to the system. These compressors are normally driven by electric motors. Interstage coolers are provided to remove heat of compression between each stage of compressor.
- ii. The total capacity of the Main Air Compressors shall be sized to design capacity, and N+1 compressors shall be specified where N is the number of compressors needed to achieve the design capacity. The spare "+1" compressor of equal size/capacity shall be installed as backup in the event of a compressor failure. All MACs shall be identical for consistency of operation and standardization of spare parts. If compressors are to be added in phases, the design shall allow for installation of additional compressors with no interruption to the fab, and N+1 compressor configuration shall be maintained at each phase of operation.
- iii. Centrifugal type main air compressors are preferred. Centrifugal compressors do not introduce lubricants into the process gas, and their design permits a more positive separation of oil-lubricated bearings from the process gas stream.
- iv. Oil-lubricated reciprocating compressors are not preferred. These can introduce hydrocarbon contaminants into the air separation process which may accumulate in the oxygen-rich liquid phase in the low-pressure distillation column.

Historically, this accumulation of hydrocarbons has been attributed to causing explosions in the cold box.

- v. Noise limit from rotating equipment and gas plant equipment (e.g. cooling tower) and piping shall meet local requirement. Additional simulation and noise attenuation to be considered and included to meet the requirement. Additional measures and accessories to be provided during the commissioning phase if the noise limit exceeding the local requirement. Gas supplier to guarantee the noise limit meeting local code and legal requirement; during operation, maintenance, and plant upset.
- vi. Supplier shall also consider the following in the selection of the Main Air Compressor:
 - Manufacturer's standard size offering the highest energy efficiency and/or lowest capital cost;
 - Gas plant is commissioned to produce 50%, 80% and 100% of individual gas plant capacity including N+1 MAC configuration to increase flexibility and reliability.
 - Existing compressors on site or nearest Micron fab to allow for sharing of spare parts in the event of an emergency;
 - Space required to accommodate all compressors needed at full buildout.
- vii. Where 2 or more air separation plants are planned to be built next to each other, the main air compressors should be shared including the spare "+1" compressor. The main air compressors are to be connected or manifolded to a common outlet air header such that combinations of various MACs can be run to provide optimum (minimum) power consumption at any stage of demand ramp up.
- viii. Two independent power sources/supplies shall be provided to the gas plant supplier. Segregation by power source should be applied accordingly to the N+1 major equipment to reduce the shutdown impact due to utility interruption.
- ix. Safeguard Main Air Compressors in accordance with FM Data Sheet 7-95, Compressors.

5.3.3 Pre purification unit

- i. The Pre-Purifier Unit (PPU) removes moisture, CO₂, and most hydrocarbons from the air. Moisture and CO₂ must be removed to prevent ice and dry ice from forming later in the process. A PPU is typically made up of a chiller to cool the air to 40-55 °F, a condensate separator to remove free water and 2 vessels filled with a combination of alumina and molecular sieve material, which adsorbs the contaminants while allowing the air to pass through. Molecular sieve beds need regular regeneration using heated waste nitrogen to remove accumulated contaminants. There are usually two parallel beds: one on line, one regenerating. Beds switch every 5-8 hours for regeneration. The air exiting from the PPU is very close to moisture and CO₂ free.
- ii. The supplier shall recommend the size and capacity of the pre-purification system as appropriate for the size of the gas plant to be constructed. Since a PPU typically consists of a chiller, a condensate separator, and 2 vessels/beds filled with a combination of alumina and molecular sieve material, the supplier shall consider the failure mode, probability of failure, and mean-time to repair these components and recommend if an "N+1" configuration may be required to avoid a single point of failure. Design must allow for regeneration of the molecular sieve every 5-8 hours or as required by the supplier's standard operating procedures without interruption of gas supply to the fab.
- iii. The supplier shall also consider the following in selection of the Pre-Purification Unit:
 - Manufacturer's standard size offering the lowest capital cost.
 - Existing PPU's on site or nearest Micron fab to allow for sharing of spare parts and material in the event of an emergency.
 - The PPU shall have 2 parallel beds, at a minimum (one on line and the other in regeneration) with switching capability (automatic switching preferred).
- iv. To control the contamination level, it is best practice to monitor PPU performance by tracking CO₂ levels at the PPU outlet with a continuous analyzer. Levels of 1 ppmv of CO₂ are typically recognized as an alarm/unacceptable level. CO₂ is the first contaminant to break through and is an auxiliary indication of a need for regeneration.

5.3.4 Pressure Control Manifold

- i. Pressure control valves will be necessary to achieve the performance specifications as required by Micron. The supplier shall select and size the pressure control manifold and valves based upon the total flow at full ramp. The manifold shall be designed for ease of replacement and periodical maintenance or repair of a pressure control valve without causing a supply interruption to the fab. An "N+1" configuration of pressure control valves should be maintained.

5.3.5 Heat Exchanger

- i. Exchanger will be necessary to recover the coldness of products in cold box and in turn cool down the air feed after the compressor. The entering air through heat exchanger will be cooled in the counter flow of the gas products coming off from the cold boxes
- ii. Good heat insulation shall be required and typically they are confined into metal boxes filled with perlite (or rockwool for older model)
- iii. Heat/cold energy balances calculation and modeling shall be provided to Micron as part of the assessment for energy sustainability and conservation.
- iv. For plate/fin type heat exchanger, hazardous amounts of hydrocarbons can accumulate between much smaller fluid flow passage and precipitate out even though overall accumulations remain well above dangerous levels. The plugged passage could also lead to boiling without replenishment (dry boiling) and eventual dry spots. Preventive measures to control and remove the accumulations (mainly CO2 and water) along with other design features shall be considered to limit the formation of plugs and the dead-end boiling. Preventive maintenance plan to be developed and maintained to ensure operational reliability
- v. Reference: FM Data Sheet 7-35, Section 3.1.3 Heat Exchangers

5.3.6 Cold boxes

- i. The Cold Box contains the cryogenic heat exchangers, distillation columns and associated valves and piping. Because parts of this system are very cold, all components are mounted inside the cold box and then encased in insulation. Cold boxes can be rectangular or cylindrical and are usually tall, some over 200 ft depending on capacity.

- ii. Modern cold boxes are typically filled with perlite insulation. Older cold boxes may be tightly packed with cryogenic rockwool.
- iii. The supplier shall size the capacity of the cold box as appropriate for the size of the gas plant under construction. The supplier shall consider the failure mode, probability of failure, impact of failure, and mean-time to repair components within the cold box and recommend if an "N+1" configuration is necessary to avoid a single point of failure.
- iv. Provide a continuous purge of dry nitrogen for the cold box to prevent entry of outside air. Water vapor in the air could cause ice that diminishes the thermal properties of the insulation and result in increased stress on internal pipe and fittings. Air could also be liquefied by the cold surfaces of process equipment giving an oxygen-enriched space. Any ignitable liquid or combustible (e.g., oil leaks) could be more readily ignited.
- v. Frost accumulation on the cold box is an indication of poor insulation or failure of equipment containing cryogenic liquid. These leaks could cause structural failure (from extreme cold) or ignition or explosion if small oil leaks or combustible residues are present in the oxygen-enriched atmosphere. Materials of construction should be suitable for the expected temperatures.
- vi. Reference: FM Data Sheet 7-35, Air Separation Unit

5.3.7 Cooling tower

- i. Cooling towers supply water primarily for the main air compressors. Cooling towers are often made from combustible construction. Sprinklers may be required in cooling towers of combustible construction. Use of noncombustible designs would minimize the potential for interruption of cooling water supply. Refer to FM Data Sheet 1-6, Cooling Towers.
 - ii. Cooling tower to be designed with N+1 capacity to enable uninterrupted supply during maintenance of the cooling tower.
 - iii. The cooling tower type and design shall be an FM Approved cooling tower as listed in FM Data Sheet 1-6, Cooling Towers.
 - iv. Use galvanized steel, cementitious, or fill material FM approved for the particular tower model. Otherwise, protect the fill. Do not use polystyrene fill, even if sprinkler protection is provided.
 - v. Construct fan decks, distribution decks, louvers, and end walls of noncombustible or FM Approved material.
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Otherwise, provide sprinkler protection. Avoid the use of combustible louvers, even on protected towers, because they are not usually covered by the fire protection system and could be ignited by an exterior exposure fire.

- vi. Protect cooling towers from external fire exposure through the use of sprinkler protection or provide fire partitions of at least one-hour fire resistance between towers and structures.
- vii. Provide protection against wind forces in accordance with FM Data Sheet 1-28, Wind Design.
- viii. Provide protection against earthquakes in accordance with FM Data Sheet 1-2, Earthquakes.
- ix. Provide sprinkler protection based upon the type of cooling tower (counterflow induced-draft towers, crossflow towers with open distribution basins, etc.).
- x. Provide an adequate water supply to simultaneously supply all discharge outlets in the two longest, adjacent sprinkler systems. Include a minimum of 500 gal/min (1900 L/min) for hose streams in addition to sprinkler requirements in the design of water supplies. The water supply shall be sized for at least a 2-hour duration for sprinklers and hose stream.
- xi. Details to be referred to Global Facilities - Mechanical – Chilled Water and Condenser Water System Standard
- xii. References: FM Data Sheet 1-6, Cooling Towers; FM Data Sheet 1-28, Wind Design; FM Data Sheet 1-2, Earthquakes.

5.3.8 Expander

- i. Expanders provide the required refrigeration to produce liquids in the distillation column system (cold box). Air, Nitrogen or Waste Nitrogen is fed to the expander, causing the wheel expander to turn and transfer energy to a compressor, generator or oil brake. This transfer of energy causes the gas to cool. As the process continues, the outlet temperature of the expander eventually reaches design temperature while cooling the cold box.
- ii. The supplier shall select and size the capacity of the expander as appropriate for the size of the gas plant to be constructed. The supplier shall consider the failure mode, probability of failure, impact of failure, and mean-time to repair the expander and recommend if an "N+1" configuration is necessary to avoid a single point of failure.

5.3.9 Derime

- i. Deriming is the process of completely warming up the entire plant for the purpose of removing contaminants that may be present throughout the plant in exchangers and columns. Frequency is determined by the loss of efficiency in the exchangers or other indicators. Frequency of deriming per year is recommended, but at least one deriming per three years depending on the ASU condition [Routine Maintenance]

Abnormal events or other indicators/operation parameters deviating from regular gas plant operation may need to perform more frequently as and when as per recommendations by the supplier to ensure long-term continuous functionality of the plant.

5.3.10 Transformers

- i. Transformers are needed for providing reliable supplies of the large amounts of electric power needed. These are usually located so as not to present an exposure to, or be exposed by, the process or storage tanks.
- ii. The supplier shall perform an engineering analysis to determine an adequate electrical protective scheme for each transformer, taking into account the following factors:
 - The criticality of the transformer
 - The lead time required to replace or repair the transformer
 - The fire exposure presented by the transformer to buildings and equipment
 - The effect of transformer failure on system integrity and reliability
 - The service requirements of the transformer (e.g., is the transformer required to operate in a high fault environment, or is it exposed to high levels of harmonics, overvoltage, and lightning?)
- iii. Provide overvoltage transient protection for all transformers in accordance with FM Data Sheet 5-11, Lightning and Surge Protection for Electrical Systems.
- iv. Provide differential protection to cover all the windings on multiple-winding power transformers, including specialty transformers such as rectifier, induction, etc. when applicable.
- v. Provide overcurrent and ground fault protection for the tertiary winding of three-winding transformers and autotransformers.

- vi. Provide an arc-monitoring system to detect arcing faults in transformer vaults with exposed energized components.
- vii. Provide continuous online condition monitoring systems to monitor the following parameters:
 - Moisture-in-oil
 - Temperature
 - Dissolved gases
 - Power Factor for bushing with testing tap
 - Partial discharge activity monitoring
 - On-load tap changer motor current
- viii. The continuous online condition monitoring systems shall have the following features and capabilities:
 - The ability to either transmit data continuously to a manned location (SCADA), or the ability to allow remote retrieval of the data
 - The ability to generate alarms (to a staffed location) when unusual conditions are detected
 - The ability for the user to set caution and alarm levels
 - The ability to store and trend monitored parameters
 - Field calibration and self-diagnostic capabilities
 - A measurement accuracy of + 5% for DGA gases and moisture-in-oil
 - A measurement accuracy of + 1% for bushing leakage current monitoring and be able to accept leakage current up to 300 mA
 - A sampling rate of at least once every 4 hours (an hourly sampling rate is preferred)
 - Robustness and immunity to electromagnetic fields interference and radio frequency interference
 - Security to prevent unauthorized changes to alarm conditions and limits
- ix. Provide fire protection and detection for indoor transformers.
- x. Provide sprinkler protection as required. For FM Approved transformer fluids, provide sprinkler protection with a design of 0.2 gpm/ft² (8 mm/min) over the entire room. For transformers using non-FM Approved fluids, provide sprinkler protection in accordance with FM Data Sheet 7-32, Ignitable Liquid Operations. Also design containment and drainage in accordance with FM Data Sheet 7-32.
- xi. Provide fire protection for outdoor transformers.
- xii. Provide exposure protection for outdoor transformers.
- xiii. Details to be referred to Global Facilities System Standard Specification – Electrical System Design
- xiv. References: FM Data Sheet 5-4, Transformers; FM Data Sheet 5-11, Lightning and Surge Protection for Electrical Systems; FM Data Sheet 7-32, Ignitable Liquid Operations

5.4 Gas Plant capacity and sizing

At a minimum, the plant's capacity to produce the bulk gases (N2, O2, Ar) shall be sized for 40% more than the projected demand/forecast for each gas at full ramp (The forecast used shall be as far as possible or reasonable at a minimum of 3 years). When the bulk gases are produced by independent plants, each plant shall be sized for 40% over-capacity at a minimum. In a multiple plant configuration in which the plants are connected, the sum of all the plants capacities shall be 40% more than the total projected demand/forecast for each gas at full ramp. This is in-line with our current Business Continuation Plan (BCP); whereby an expansion and new gas plant will be reviewed and considered whenever the utilization rate is at 80%. Selection of the size, quantity, and manufacturer of the gas plant(s) shall also consider the following:

- Manufacturer's standard plant sizing and capacity offering the highest energy efficiency and/or lowest capital cost;
- Existing gas plants on site or nearest Micron fab to allow for sharing of spare parts in the event of an emergency and to optimize spare parts inventory and cost;
- Available area for location of gas plant(s)

The gas plant capacity is referring to N2, O2 and Argon capacity as well where it's applicable. Depending on economics; there will be excess or liquid supply/haul in need to be in-place to supplement the gas plant generation for production.

5.5 Backup and redundancy requirement

Gas plant shall be designed to be back-up by either;

- Standby gas plants
- Liquid storage on site

The quantity and amount of the backup shall be evaluated based on below;

Description	Parameters
On-site Storage (including off-site storage that are connected)	A
# haul in/day supplier commitment/capability	B
# haul in/day required with largest ASU failure	C
Continuous haul in committed by gas supplier (days)	D

On-site storage (including off-site storage that are connected) (A) :
A>72 hours backup duration, based on two (2) largest gas plants capacity (this is to meet the overall specific site BCP and shredding plan

with manufacturing in the event of gas supply interruption). If the inventory can't be met, mitigation plan/site management must agree with the proposed setup and proposed inventory for site BCP risk management.

Haul in : B>C (additional mitigation and provision need to be considered if this can't be met. Logistics and gas supplier BCP for Micron to be reviewed and accessed)

Continuous haul in committed by gas supplier : D>15 days (this is to meet the overall specific site BCP in the event of gas supply interruption)

Redundancy for equipment and component level will be described at component level to eliminate single point of failure (SPOF) which will impact supply of the process gas to the fab.

5.6 Typical power requirement

5.6.1 General

- MV (11kV) and LV (415V) or any other determined with gas supplier during design phase
- The electrical system shall be Selectively Coordinated, including any overcurrent devices upstream
- The starting of a compressor motor or energizing any other large load shall not create a voltage drop of more than 4% at the Point of Common Coupling.
- Total harmonic distortion of the electrical system shall not exceed 10% for current and 5% for voltage
- Dual feed/incomer to have redundancy and backup (Normal vs emergency vs UPS). Main power feed shall be 100% redundancy and distribution network should be reviewed and designed to support the downstream N+1 equipment.
- The dual power supply and dual bus system shall be designed for redundant equipment. The tie breaker shall be provided to achieve the function of power backup.
- N+1 redundancy for each equipment unit operation (exceptions to be reviewed and approved). Common controls between redundant systems powered from UPS and normal power.
- Manual bypasses and control of automated control valves need to be in-place.
- UPS supply for critical instruments and control

- Programmable Logic Controllers (PLCs) and analyzers to be backed by emergency-backed UPS
- Spare circuit breakers; large amperage breakers shall be in-place for future expansion
- Details to be referred to Global Facilities System Standard Specification – Electrical System Design

5.7 Typical utility requirement

5.7.1 General

- Provision of laydown area, site access
- Pressure vessel approval and documentation from authority
- Construction permitting
- Civil and electrical power provision and permitting
- Temporary power for construction and commissioning
- Make-up water for cooling system
- Internet connection for telemetric monitoring and etc

5.7.2 Civil works

- Soil survey, underground mapping
- Detailed civil design based on gas supplier layout
- Civil design and authority submission
- Concrete foundation for equipment
- Compound lighting
- Trenches for cables and underground piping
- Water pit for condensate discharge and perimeter drainage
- Earthing
- Perimeter fencing gates and access road
- Gravel filling for gas yard area

5.8 Continuous Quality Control (CQC) requirements

Continuous Quality Control (CQC) shall be provided and monitored in the GMS/SCADA. The parameters needed to be monitored shall be in-line with all the required specified specifications. Sampling program with sampling frequency, location, procedure, and required impurities to be analyzed, as well as the laboratory qualified for the analysis

Suppliers shall provide analyzed impurities with detection limits, CQC analyzers or analyzed samples analytical methodology, CQC features. The CQC analyzers or sample analytical methods shall be proven to have detection limits below specifications.

5.9 Signal from the plant requirement

- i. To control the contamination level, it is best practice to monitor PPU performance by tracking CO2 levels at the PPU outlet with a continuous analyzer. Levels of 1 ppmv of CO2 are typically recognized as an alarm/unacceptable level. CO2 is the first contaminant to break through and is an auxiliary indication of a need for regeneration.
- ii. Interlock and alarm for breakthrough of O2 to N2 purifier
- iii. Interlock for purifier high temperature alarm is required to prevent fire and damage to the purifier.
- iv. Interlock/alarm for low temperature gas entering purifier to prevent damage to purifier
- v. Interlocks and alarms shall be available for the quality to be determined to ensure quality downstream is meeting Micron's requirement (interlock on gas plant CQC and post purifier CQC)
- vi. All alarms indicating a potential problem that can affect the gas quality and/or the gas plant functionality shall be available for monitoring by the Micron SCADA, and be associated to an auto paging system to notify the designated supplier operation work group. Suppliers shall provide a fault alarm list to be wired to the Micron SCADA.
- vii. The main operating parameters, flow rate and pressure, shall be monitored by the Micron SCADA. Based on supplier specifications for all relevant parameters of the air separation process, flow, pressure, temperature of delivered gases, storage tank level, temperature of the ambient vaporizers.
- viii. Continuous Quality Control system (CQC) parameters shall be available for Micron SCADA monitoring.

5.10 Site / space requirement (incl. specific requirements for O2)

- i. In some specific cases (closed environment), O2 level monitoring could be required when the N2 level could reduce the O2 level below 19.5%
- ii. Quarterly reports will be issued by the supplier with all relevant CQC historical data, a power efficiency O&M summary, proposed upgrading and risk mitigation improvement, and any other item functional information.
- iii. As part of contingency plan, the supplier will provide:

Global Facilities - Design & Construction Standard - Bulk Gas - Gas Plant (N2, O2, Ar) Design System

Last Modified: 10/18/22

Version: 2

ID: A3YRXSD74VDV-57553043-381

- Maximum number of tankers/trucks (loads) that supplier can ensure on a daily basis and each tanker LN2 and LO2 capacity. Please provide total commitment in both tanker loads and tons per day.
- Any limiting factor in logistics and transportation.
- The local/country LN2 and LO2 inventory (LN2 and LO2 storage capacities and location).
- The number of available plants and distance from Micron site.
- The capability to employ additional resources from other sites and/or plant locations, and the amount of time needed to mobilize those resources.
- The quality assurance (by CoA or IQC) of offloaded LN2, LO2 and LAr to prevent quality excursions in an emergency situation.
- The list of critical spare parts available on site, and/or in the proximity hub/warehouse. For not available materials, the supplier will notify the longest lead item to get available on site.
- Number of tanker loads and tons per day required to support a one plant down unplanned event, per supplier's proposal gas plant size.
- Supplier will be required to review/re-evaluate the contingency plan on regular basis and/or depending on facility and manufacturing expansion.
- The future LN2 and LO2 storage capacity shall be identified according to the Supplier's historical failure data, contingency plan, and as a result of a risk mitigation analysis.
- Contractor shall provide a third-party QA to monitor installation methods, certify incoming material, validate system integrity, and aid in system startup and quality monitoring. The QA/QC personnel proposed by contractor must be reviewed and approved by Owner before beginning work on Project.

5.11 Preferred gas plant suppliers

Gas Plant Suppliers: Air Liquide, Air Products, Linde, Air Water, Praxair

5.12 Excerpt and reference with relevant code and requirement

5.12.1 Industry Safety Requirements (NFPA, FM Global, IFC, etc.)

The gas plant shall be designed according to relevant local codes and regulations, including but not limited to International Fire Code (IFC) and local/country fire regulations, International building code (IBC), Safety and Health Administration (OSHA),

Environmental Protection Agency (EPA), and Factory Mutual (FM) Data Sheets (7-35 and 7-35R, Air Separation Processes).

Preliminary design shall be submitted to Factory Mutual (FM Global). FM Global's recommendations shall be evaluated during the design phase.

The ASU and all other plant components and sub-assemblies shall be designed by the gas company (the supplier) using state-of-the-art air separation technology and energy conservation.

Design vessels in accordance with the requirements of the ASME (American Society of Mechanical Engineers) Boiler and Pressure Vessel Code, Section VIII – Unfired Pressure Vessels, or international equivalent.

Fabricate piping or tubing from materials meeting the requirements of ANSI/ASME B31.3, Code for Chemical Plant and Petroleum Refinery Piping or international equivalent.

Route waste streams that are oxygen enriched or have low oxygen content to a safe location. Oxygen enriched streams present a hazard of accelerated combustion to ordinary air, while oxygen deficient streams could create a personnel hazard (asphyxiation).

Use gaskets, lubricants, piping, etc. that have been tested as suitable for oxygen service.

Safeguard air, nitrogen, and oxygen compressors in accordance with FM Data Sheet 7-95, Compressors.

Wherever cryogenic liquid can be trapped between closed valves, provide pressure relief valves designed to relieve the pressures generated by the trapped liquid as it warms via exterior heat "leaks". They need not be designed based on process liquid flows.

Depending on the suppliers' technology and projected demand, the ASU shall be designed to produce high purity N2 (HPN2) or general N2 (UN2) grade with dedicated N2 purifiers.

The Suppliers shall provide a "cost of ownership" evaluation of the gas plant including: the power consumption per unit volume of N2 produced, cost/benefit comparison, O&M requirements, and total cost of ownership of the following plant features of each proposed option:

- i. HPN2 versus UN2 air separation system performance, energy requirements, and purifiers features
- ii. Liquid nitrogen (LN2) and/or ultra high purity (UHP LN2) production, capacity, and storage versus total gaseous N2 production

Liquid product (LN2, LO2, and LAr) shall be used as the primary backup in the event of a planned and/or unplanned plant shutdown. The liquid products shall be stored on site in cryogenic tank(s). PN2, PO2, and PAr backup shall guarantee that the purifiers meet the maximum purified gas demand and specification. General purity (i.e., UN2) and purified gas (i.e., PN2) supplies shall be guaranteed by vaporized liquid products. Refer to "Bulk Gas Storage and Distribution System Design and Performance Requirements" for additional requirements.

Supplier shall perform a Design Hazard Review to ensure that all hazards are addressed. The DHR shall use the HAZOP technique, or similar, to provide a structured and documented process to ensure that all aspects of the plant are addressed, including startup, shutdown, normal operation, materials of construction, etc. The supplier shall review the DHR with Micron as part of the design review and approval process.

Supplier shall retain a consulting firm specializing in earthquake design and evaluation to review all aspects of facility construction, process service equipment, and building service equipment for local code compliance and compliance with FM Global Data Sheet 1-2.

Use an importance factor of "Critical Facility" for the entire site per Data Sheet 1-2.

Design anchoring and bracing of equipment to resist a minimum horizontal acceleration of 0.5 g. If the applicable recommendations of Data Sheet 1-2 Earthquakes or the local code specify a larger g-factor, use that.

Supplier shall provide signed and stamped original seismic calculations and designs from the consulting firm for Micron's review and approval and submission to FM Global.

Reference: FM Global Data Sheet 17-12, Semiconductor Fabrication Facilities, 2.2.15 Earthquake

Prior to starting up the plant, a formal, documented Operational Readiness Inspection, or equivalent, is performed to ensure that all aspects of the DHR have been incorporated and that the plant is ready for operation. The supplier shall provide a copy of the ORI report to Micron for review and acceptance prior to starting up the plant.

References: FM Data Sheets 7-35 and 7-35R, Air Separation Processes; 7-95, Compressors; 12-43, Pressure Relief Devices

Global Facilities - Design & Construction Standard - Bulk Gas - Gas Plant (N2, O2, Ar) Design System

Last Modified: 10/18/22

Version: 2

ID: A3YRXSD74VDV-57553043-381

6.0 Document Control

6.1 **Approval:** GLOBAL_FAC_GFTT_DOC_APPR

6.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GFTT LEADS
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

6.3 Final Viewer Access

All Micron Team Members from all Micron Sites

6.4 Retention

Adherence to the requirements specified in the
[**Global Facilities – Document Control Procedures**](#)

6.5 Review

Biennially (two years)

Global Facilities - Design & Construction Standard - Bulk Gas - Gas Plant (N2, O2, Ar) Design System

Last Modified: 10/18/22

Version: 2

ID: A3YRXSD74VDV-57553043-381

7.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	07/14/20
1	5.0	5.1 Added "the gases are vented" 5.3.1 Added "Upstream Pre-" and "Post gas plant" to i and ii respectively 5.3.5 Added "Heat Exchanger" 5.3.9 Added "Frequency of deriming per year is recommended, but at least one deriming per three years depending on the ASU condition [Routine Maintenance] Abnormal events or other indicators/operation parameters deviating from regular gas plant operation may need to perform more frequently as and when as per recommendations by the supplier to ensure long-term continuous functionality of the plant." " 5.5 Added "including off-site storage that are connected" Added "(including off-site storage that are connected) (A) : A>72 hours backup duration, based on two (2) largest gas plants capacity" Added "If the inventory can't be met, mitigation plan/site management must agree with the proposed setup and proposed inventory for site BCP risk management."	10/18/22

Global Facilities - Design & Construction Standard - Bulk Gas - Gas Plant (N2, O2, Ar) Design System

Last Modified: 10/18/2022

Version: 2

ID: A3YRXSD74VDV-57553043-381

Appendix 1: Template Form for Site Specific Deviation List

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT